

Methods: Analysis of Feeding Behavior in Mice

Methods

Data Extraction and Preprocessing

Raw feeding event data were extracted from FED3 devices, which recorded timestamped pellet retrieval events. The data included the time of each pellet retrieval, and inter-pellet intervals (IPIs) were calculated as the time difference between consecutive pellets. The data were preprocessed to convert timestamps into hours relative to a reference point (e.g., lights on) and filtered to include only relevant events. This preprocessing step ensured that the data were in a consistent format for downstream analysis.

Kernel Density Estimation (KDE) of Inter-Pellet Intervals

To characterize the distribution of inter-pellet intervals (IPIs), a kernel density estimation (KDE) was performed. The KDE curve was computed using the log-transformed IPIs (in minutes) to visualize the probability density of intervals. The KDE was defined as:

$$f(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - \log_{10}(\text{IPI}_i)}{h}\right),$$

where:

- $f(x)$: Probability density at x , representing the likelihood of observing an IPI of x minutes.
- n : Number of IPIs in the dataset.
- h : Bandwidth parameter, controlling the smoothness of the KDE curve.
- K : Kernel function (e.g., Gaussian), which determines the shape of the distribution around each data point.
- IPI_i : Inter-pellet interval for the i -th event, log-transformed to better visualize the wide range of intervals.

The KDE curve revealed that most IPIs were short, with a sharp peak around 1 minute, indicating that mice typically take pellets in rapid succession. The curve also showed a long tail, representing less frequent longer intervals.

Cluster Size Distribution

Feeding events were grouped into clusters based on a temporal threshold of 60 seconds. A cluster was defined as a sequence of pellets where the IPI between consecutive pellets was less than or equal to 60 seconds. The distribution of cluster sizes (number of pellets per cluster) was computed, and the average number of events for each cluster size (1 pellet, 2 pellets, etc.) was calculated across all mice. This analysis revealed that:

- Single-pellet events were the most frequent.
- The frequency of clusters decreased as the cluster size increased.
- Clusters of 2-5 pellets were relatively common, while larger clusters (6+ pellets) were rare.

Exponential Decay Model

The distribution of cluster sizes was fitted to an exponential decay model to describe the relationship between cluster size and frequency. The exponential decay model was defined as:

$$y = a \cdot e^{-bx},$$

where:

- y : Average number of events for a cluster of size x .
- a : Initial amplitude, representing the expected count for a cluster size of 1.
- b : Decay rate, indicating how quickly the counts decrease with increasing cluster size.
- x : Cluster size (number of pellets in the cluster).

The fitted parameters were $a = 139.55$ and $b = 0.28$, with an R^2 value of 0.9668, indicating a strong fit. This model confirmed that the frequency of clusters decreases exponentially with increasing cluster size, consistent with the observed data.

Conditional Probability Analysis

The conditional probability $P(x + 1|x)$ was computed to determine the likelihood of adding an extra pellet to a cluster of size x . This probability was calculated as:

$$P(x + 1|x) = \frac{N_{x+1}}{N_x},$$

where:

- N_x : Number of clusters of size x .
- N_{x+1} : Number of clusters of size $x + 1$.

The results showed that:

- $P(2|1) = 0.6290$: There is a 62.9% chance of adding a second pellet to a single-pellet event.
- $P(3|2) = 0.9774$: There is a 97.7% chance of adding a third pellet to a two-pellet cluster.
- The probabilities decreased for larger cluster sizes, with a significant drop at $x = 5$.

Delta P: Changes in Conditional Probability

The change in conditional probability, $\Delta P(x)$, was calculated as the difference between consecutive probabilities:

$$\Delta P(x) = P(x + 1|x) - P(x|x - 1).$$

The values of $\Delta P(x)$ were:

- $\Delta P(1) = 0.3484$: A large positive change, indicating a significant increase in the likelihood of adding a second pellet after a single-pellet event.
- $\Delta P(2) = -0.1872$: A negative change, indicating a decrease in the likelihood of adding a third pellet after a two-pellet cluster.
- $\Delta P(5) = 0.2315$: A positive change, indicating an increase in the likelihood of adding a sixth pellet after a five-pellet cluster.

The positive $\Delta P(5)$ suggests that mice are more likely to add a sixth pellet after a five-pellet cluster compared to adding a fifth pellet after a four-pellet cluster. However, the overall trend shows that the likelihood of adding another pellet decreases as the cluster size grows, with the largest drop occurring at $x = 5$. This drop indicates that clusters of size 5 are less likely to grow further, making the 5th pellet a natural boundary for defining meals.

Wilson Confidence Intervals

To assess the reliability of the conditional probabilities, 95% confidence intervals (CIs) were computed using the Wilson score interval method. The Wilson interval is a robust method for estimating confidence intervals for binomial proportions, especially when sample sizes are small or probabilities are close to 0 or 1. The confidence interval for $P(x + 1|x)$ was calculated as:

$$CI = \hat{p} \pm z \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}},$$

where:

- $\hat{p}$: Observed probability $P(x + 1|x)$.
- z : Z-score corresponding to the desired confidence level. For a 95% confidence interval, $z = 1.96$, which corresponds to the 97.5th percentile of the standard normal distribution (since the interval is two-tailed).
- n : Number of trials (clusters of size x).

The Z-score of 1.96 is used because it captures 95% of the area under the standard normal distribution, ensuring that the interval is wide enough to account for sampling variability. The Wilson confidence intervals confirmed the robustness of the conditional probabilities, with narrow intervals for most cluster sizes. For example:

- $P(2|1)$: 95% CI = [0.6012, 0.6559].
- $P(3|2)$: 95% CI = [0.9640, 0.9858].

These intervals indicate that the observed probabilities are statistically significant and not due to random variation.

Refinement of Meal Definitions

Using the determined threshold from the conditional probability analysis, feeding events were reclassified into meals, snacks, and mega meals. A meal was defined as a cluster of pellets where the conditional probability of adding another pellet fell below the threshold. This refinement allowed for a more accurate categorization of feeding behavior based on both temporal and probabilistic criteria. The results showed that:

- Meals were typically clusters of 2-5 pellets.
- Snacks were single-pellet events.
- Mega meals were rare and consisted of clusters with 6 or more pellets.

Statistical Validation

All results were validated using statistical methods, including the exponential decay model fit (R^2) and Wilson confidence intervals. This ensured the robustness and reliability of the findings. The high R^2 value (0.9668) for the exponential decay model and the narrow confidence intervals for the conditional probabilities confirmed that the observed patterns were statistically significant and not due to random variation.